

Product Requirements Document (PRD)

Fast Food Restaurant Management System (RMS)

Field	Value
Product Name	(TBD — working title: "FlavorFlow RMS")
Document Version	1.0
Date	April 2026
Status	Draft for Review
Document Owner	Product Team

1. Executive Summary

This document defines the requirements for an end-to-end, cloud-based **Fast Food Restaurant Management System**. The platform replaces the chaotic mix of paper slips, verbal orders, and disconnected apps that most quick-service restaurants rely on today. It connects five operational surfaces — **customer, waiter, reception/cashier, kitchen, and admin** — around a single source of truth for orders, inventory, and revenue.

Customers scan a QR code at their table, browse a live digital menu, place and track their order in real time, pay digitally, receive a bill on their device, and are invited to review the restaurant. Staff get purpose-built interfaces (Kitchen Display, Waiter App, Reception Terminal, Admin Console) that eliminate miscommunication and give managers live visibility into sales, food cost, wastage, and stock.

2. Vision & Goals

2.1 Product Vision

To become the operating system for modern quick-service restaurants — where every order, ingredient, and customer interaction is tracked, optimized, and turned into insight, while keeping the guest experience fast, delightful, and frictionless.

2.2 Business Goals

1. **Reduce order errors** by eliminating verbal/handwritten order flow (target: <1% error rate).
2. **Decrease average order-to-serve time** by 25% through kitchen workflow optimization.
3. **Cut food waste** by 20% via recipe-linked inventory deduction and spoilage tracking.
4. **Increase average order value (AOV)** by 15% through digital upsell, combos, and modifiers.
5. **Improve Google review volume** by 3x through in-app review prompts.
6. **Enable data-driven decisions** via real-time dashboards on sales, cost, and staff productivity.

2.3 Success Metrics (KPIs)

Category	Metric	Target
Operations	Average order-to-serve time	≤ 18 min
Operations	Order accuracy	≥ 99%
Customer	Digital order adoption rate	≥ 70% of dine-in orders
Customer	CSAT / in-app rating	≥ 4.5 / 5
Customer	Google review conversion	≥ 15% of served customers
Financial	Food cost %	≤ 32% of revenue
Financial	Wastage %	≤ 3% of inventory value
Financial	Table turnover rate	+20% vs baseline
Staff	Time to accept order in kitchen	≤ 60 seconds

3. Problem Statement

Quick-service and casual restaurants today typically struggle with:

- **Manual order chaos** — waiters forgetting items, illegible handwriting, kitchen misreads.
- **No real-time ETA** — customers have no idea when their food is coming; staff can't answer honestly.

- **Disconnected inventory** — a sold burger doesn't automatically deduct a bun, patty, and cheese slice, so stock-outs happen mid-service.
- **Wastage blind spots** — food thrown away or "staff meals" are never logged, distorting actual food cost.
- **Lost reviews** — satisfied customers leave without being asked to rate; only angry ones post.
- **Owner in the dark** — no single dashboard for today's sales, top items, wastage, and profit margin.

This system solves all of the above with one integrated platform.

4. Target Users & Personas

4.1 Primary Personas

Persona 1 — Ayesha, the Walk-in Customer (25) Wants to scan, browse pictures, customize her order, pay without waiting, and know when her food is coming. Will leave a Google review if asked at the right moment.

Persona 2 — Bilal, the Waiter (22) Moves between 6–10 tables. Needs to know instantly which table's order is ready, who called him, and which tables need cleaning. Doesn't want to write anything down.

Persona 3 — Hina, the Receptionist/Cashier (28) Acts as the floor controller — confirms orders, handles phone/takeaway orders, processes payments, resolves disputes, prints bills and KOTs.

Persona 4 — Kashif, the Kitchen Head (35) Runs the line. Needs a clear order queue, ability to set prep times, flag out-of-stock items, and mark dishes ready for pickup.

Persona 5 — Mr. Rehman, the Owner/Admin (45) Cares about daily revenue, food cost, wastage, top-selling items, and which staff member drove the most sales. Checks the dashboard on his phone multiple times a day.

4.2 Secondary Personas

- **Delivery Rider** (in-house or third-party)
 - **Inventory/Procurement Manager** (for multi-outlet chains)
 - **Accountant** (month-end reporting, tax exports)
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5. User Roles & Permissions Matrix

Role	Menu	Orders	Kitchen	Inventory	Staff	Reports	Finance	Settings
Customer	View	Create/Track own	—	—	—	—	Pay own bill	—
Waiter	View	View assigned	View ready	—	—	—	—	—
Receptionist	View	Full CRUD	View	View	—	Daily sales	Process payment	—
Kitchen Staff	View	View queue	Full	Mark depleted	—	—	—	—
Manager	Edit	Full	Full	Full	View	Full	View	Limited
Admin/Owner	Full	Full	Full	Full	Full	Full	Full	Full
Accountant	View	View	—	View cost	—	Financial	Full export	—

Role permissions must be **configurable** — an owner should be able to grant a senior waiter partial admin rights without code changes.

6. System Modules Overview

The platform is composed of **10 functional modules**:

1. Customer Digital Ordering (web app via QR)
2. Table & QR Management
3. Reception / Cashier Terminal
4. Kitchen Display System (KDS)
5. Waiter Mobile App
6. Menu & Recipe Management
7. Inventory, Procurement & Wastage
8. Staff & Shift Management
9. Customer CRM, Loyalty & Marketing
10. Reporting, Analytics & Admin Console

Each is detailed below.

7. Functional Requirements

7.1 Module: Customer Digital Ordering

Access: Via QR code scan at the table — no app install. Progressive Web App (PWA).

Features:

- **QR-linked session** — scanning auto-identifies the table, outlet, and starts a guest session.
- **Language toggle** — at minimum English + Urdu; architecture must support additional locales.
- **Menu browsing** with:
 - Categories, search, filters (veg/non-veg, spicy, bestseller, new, under Rs. X).
 - High-res photos, description, allergen tags, calorie info (optional).
 - "Customer favorites" and "Chef's picks" highlighting.
 - Real-time availability (items auto-hide when out of stock or outside time window).
- **Item customization** — size, spice level, add-ons, remove ingredients, special instructions.
- **Combos & upsells** — "Make it a meal", "Frequently ordered together".
- **Cart** — edit quantities, notes, coupon/promo code field.
- **Place order** — confirmation screen with estimated prep time.
- **Live order tracking** — status stages: *Placed* → *Confirmed* → *In Kitchen* → *Preparing* → *Ready* → *Served*. Shows live ETA countdown.
- **Add to existing order** — same session, new items go to same table bill; kitchen receives as an addendum.
- **Call Waiter / Request Water / Request Bill** buttons.
- **Split bill** option — split evenly among N guests or per item.
- **Payment** — cash (mark pay-at-counter), card (via terminal), digital wallet (JazzCash, Easypaisa, SadaPay, Stripe for international cards), bank transfer. Tipping UI.
- **Digital receipt** — shown on screen, emailed, and optionally SMS'd.
- **Feedback & review prompt** — 1–5 star + free text, then if ≥ 4 stars, **one-tap Google review deep link** with pre-filled sentiment. Below 4 stars, route to private feedback form to prevent public negative reviews and allow recovery.
- **Loyalty enrollment** — optional phone-number-based signup; points accrue automatically.
- **Order again / Recent orders** for returning customers.
- **Dietary profile** — once set, surface allergen warnings on every item.
- **Accessibility** — WCAG 2.1 AA, screen-reader compatible, high-contrast mode.

7.2 Module: Table & QR Management

Features:

- Visual floor plan editor (drag-and-drop tables, zones, sections).
 - Each table has: unique ID, capacity, zone (indoor/outdoor/VIP), status (Free / Occupied / Reserved / Cleaning / Out of Service).
 - **QR code generation** — printable, per-table, with branded template; can be regenerated if stolen/damaged.
 - **Reservations** — phone/online booking, time slots, deposit handling, waitlist.
 - **Walk-in management** — assign table manually from reception.
 - **Merge tables** (party of 10 joining two tables) and **transfer table** (move guests mid-meal) — all open orders move with them.
 - **Turn-time tracking** — how long each table takes to turn, surfaced in reports.
 - **Live floor view** for managers and receptionists (color-coded by status).
 - **Table-less modes** — Takeaway, Delivery, Dine-in via counter (no table).
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7.3 Module: Reception / Cashier Terminal

Features:

- **Live order queue** — all incoming orders (QR, phone, walk-in, delivery) in one list, color-coded by channel.
- **Order confirmation step** — receptionist reviews order, can edit, then forwards to kitchen.
- **Print Kitchen Order Ticket (KOT)** — to designated kitchen printer(s); supports station-wise routing (e.g., drinks KOT to bar printer).
- **Takeaway & phone order entry** — manual order entry UI mirroring the customer menu.
- **Delivery order entry & dispatch** — assign rider, capture customer address, print packing slip.
- **Hold / resume order.**
- **Order modifications & cancellations** — with reason codes and manager-approval threshold (e.g., cancellations over Rs. 2000 require manager PIN).
- **Bill generation & printing** — itemized, with tax breakdown, discount, tip, payment mode.
- **Split payment** across multiple methods (e.g., Rs. 2000 cash + Rs. 3000 card).
- **Refund & void** with audit trail.
- **Cash drawer management** — opening float, closing count, variance report, end-of-shift handover.
- **Shift reports** — each receptionist can close their shift with a Z-report.
- **Customer lookup** — find returning customer by phone, pull history/loyalty.

7.4 Module: Kitchen Display System (KDS)

Features:

- Tablet-first UI (big tap targets, color-coded tickets).
- **Order ticket** shows: table/order #, channel, items, modifiers, special instructions, elapsed time, priority flag.
- **Accept / Reject order** — rejection requires reason (e.g., out of stock), auto-notifies receptionist and customer.
- **Set estimated prep time** per order on acceptance — propagated to customer's tracking screen.
- **Station routing** — tickets can auto-split by station (grill, fryer, cold prep, drinks) and each station sees only its items.
- **Status updates per item** — *Queued* → *In Progress* → *Ready*.
- **Mark full order as Ready** triggers waiter notification.
- **"Bump" and "Recall"** — standard KDS actions to complete or reopen a ticket.
- **Delay / rush flag** — manually bump priority for VIPs or overdue orders.
- **86-list (out of stock)** — kitchen can flag an item as unavailable, which instantly hides it from the customer menu and alerts management.
- **Recipe card overlay** — tap an item to see the recipe (portion sizes, plating notes, allergen flags) — helps new cooks and enforces consistency.
- **Audio alerts** — chime on new order, escalating alert on overdue tickets.
- **Order aging SLA** — tickets over X minutes turn red; over Y minutes alert manager.

7.5 Module: Waiter Mobile App

Features:

- Login, see assigned section/tables.
- **Live table grid** — status, current order, elapsed time since seated.
- Push notifications: *Order ready for table 7*, *Table 3 called you*, *Table 9 requested bill*.
- **Take order on behalf of guest** (for non-tech customers or elderly guests).
- Mark order as **Picked up** → **Served**.
- Table status updates (mark cleaning done → free).
- **Tip attribution** — so waiter earnings are tracked.
- **Performance view** — tables served today, upsell count, average tip.

7.6 Module: Menu & Recipe Management

Menu Features:

- **Hierarchy:** Categories → Sub-categories → Items → Variants → Modifiers.
- Per-item fields: name (multi-lang), description, photo(s), price, tax class, SKU, preparation station, cooking time estimate, calorie/nutrition info, allergens, tags (spicy, veg, bestseller, new), availability schedule (e.g., breakfast 7–11 am only), day-of-week availability.
- **Variants** (e.g., Small/Medium/Large with different prices) and **Modifiers** (add cheese +Rs.50, extra spicy +Rs.0).
- **Combo builder** — bundle items with a bundle price.
- **Dynamic pricing** — happy hour, weekend surcharge, delivery markup.
- **Multi-outlet support** — menu can differ per outlet; central template with local overrides.
- **Bulk import/export** via CSV.
- **Version history & rollback** — every menu edit is auditable.
- **A/B test menu layouts** (future phase).

Recipe Management Features:

- For each menu item (and each variant), define the **Bill of Materials (BOM)**: every raw ingredient, quantity, and unit.
 - Support nested recipes — a "Burger Bun" can itself be a sub-recipe with flour/yeast/etc.
 - Standardized units with auto-conversion (g/kg, ml/l, piece, etc.).
 - **Portion cost auto-calculation** — sum of ingredient costs = plate cost; system shows margin vs. sale price.
 - **Yield management** — a 1 kg dough yields 8 buns; the system handles the math.
 - **Allergen aggregation** — allergens roll up from ingredients to the final item automatically.
 - **Recipe cards** — printable, displayable on KDS.
 - **Theoretical vs actual** consumption — compare what *should* have been used (by sales) vs what *was* used (by stock count); highlight variance as potential wastage/theft.
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7.7 Module: Inventory, Procurement & Wastage

Inventory Features:

- Real-time stock levels per ingredient per outlet.
- **Auto-deduction** — every sold order deducts ingredients via recipe BOM.
- Units, conversions, packaging (case → kg → g).
- **Par levels & reorder points** — alert when stock hits minimum.
- **Expiry date & batch/lot tracking** — FEFO (First Expired First Out) suggestions.
- **Stock takes** — scheduled physical counts with variance report.
- **Transfers between outlets** (for chains).
- **Cost tracking** — FIFO or weighted average cost method, configurable.

- **Low stock & expiry dashboard** with color-coded alerts.

Procurement Features:

- **Supplier database** — contacts, items supplied, lead time, price history.
- **Purchase orders** — raise PO, approval workflow, email to supplier, track status (Draft → Sent → Partially Received → Received → Closed).
- **Goods received note (GRN)** — receive stock against PO, capture actual quantity, price variance, quality notes.
- **Supplier performance** — on-time rate, price trend, quality rating.

Wastage Features:

- **Manual waste log** — any staff can record: item, quantity, reason (spoiled, dropped, overcooked, customer return, staff meal, complimentary, shift end discard).
 - Photo evidence optional but encouraged.
 - Manager approval for wastage above threshold.
 - **Wastage dashboard** — by item, by reason, by shift, by staff member — to surface patterns.
 - Variance report (theoretical consumption vs actual) automatically flags hidden wastage.
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7.8 Module: Staff & Shift Management

Features:

- Staff profile: role, contact, CNIC/ID, hire date, hourly rate / salary, bank details, emergency contact.
 - **Shift scheduling** — weekly roster view, drag-drop shifts, shift swap requests, publish and notify.
 - **Attendance** — clock-in/out via PIN, biometric, or QR; geofenced if required.
 - **Leave management** — request, approve, track balance.
 - **Performance metrics** — orders served, upsell rate, average tip, customer ratings, punctuality.
 - **Training & certifications** — food safety, hygiene expiry reminders.
 - **Payroll export** — hours × rate + tips + bonuses, CSV export to payroll provider or accounting software.
 - **Incident log** — complaints, warnings, accolades.
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7.9 Module: Customer CRM, Loyalty & Marketing

Customer CRM:

- Unified profile: name, phone, email, birthday, anniversary, dietary preferences, allergies, favorite items, lifetime value, visit frequency, last visit.
- **Automatic profile creation** on first digital order.
- Order history with re-order button.
- Notes (e.g., "always wants extra napkins").

Loyalty Program:

- Points per Rs. spent, configurable earn rate.
- Redemption — points for discount, free item, tier upgrade.
- **Tiers** (Bronze/Silver/Gold) with escalating perks.
- Birthday/anniversary auto-reward.
- Referral program — refer a friend, both get bonus.

Marketing:

- **Promo codes & coupons** — percentage, flat, BOGO, free item, first order, minimum basket.
- **Campaigns** — bulk SMS/email/WhatsApp push. Segmentation (e.g., "customers who haven't visited in 30 days").
- **Happy hour rules.**
- **Feedback & reviews dashboard** — all ratings and comments in one place, reply to public Google reviews, track sentiment over time.
- **Negative feedback alert** — any rating ≤ 3 stars triggers an instant manager notification for service recovery.

7.10 Module: Delivery & Takeaway

Features:

- **Takeaway queue** — customer places order via QR-less link or phone, picks up in person. SMS when ready.
- **Delivery orders** — address input with map pin, delivery fee by zone, minimum order value per zone.
- **Rider management** — in-house riders with profile, live location, current assignments.
- **Dispatch view** — map of active deliveries, assign riders.
- **Customer tracking link** — shared with delivery customer via SMS/WhatsApp, shows rider location.
- **Third-party integrations** — Foodpanda, Uber Eats, Careem, etc. — orders appear in the same queue via APIs/webhooks (where available), or at minimum a tablet-aggregation bridge.
- **Delivery KPI tracking** — avg delivery time, on-time %, failed deliveries, rider efficiency.

7.11 Module: Reporting, Analytics & Admin Console

Dashboards:

- **Today** — live revenue, orders, AOV, top items, pending orders, table occupancy.
- **Sales** — hourly, daily, weekly, monthly, YoY; by channel (dine-in/takeaway/delivery); by outlet; by payment mode.
- **Menu performance** — best/worst sellers, profit per item, menu engineering matrix (Stars/Plowhorses/Puzzles/Dogs).
- **Food cost & margin** — plate cost vs sale price, food cost %, theoretical vs actual variance.
- **Wastage** — trends, causes, cost impact.
- **Inventory** — current value, stock turn rate, dead stock, soon-to-expire.
- **Staff** — sales per waiter, shift productivity, tip distribution, attendance.
- **Customer** — new vs returning, retention cohort, LTV, top spenders.
- **Tables** — turn time, revenue per table.
- **Reviews & CSAT** — rating trend, response rate, negative feedback funnel.

Operations:

- Exportable reports (PDF, CSV, Excel).
- Scheduled email reports (daily/weekly digest to owner).
- **Drill-down** from any chart to underlying transactions.
- Custom date ranges and comparisons.
- **Anomaly alerts** (e.g., "Sales today are 40% below same day last week").

Admin Console also covers:

- Global settings: business info, tax rules, service charge, tipping, currency, timezone, receipt templates.
- Printer setup (per station).
- Payment gateway configuration.
- Notification templates (SMS, email, push) with variables.
- Audit log — every sensitive action (price change, void, refund, permission change) logged with user, timestamp, before/after.
- Multi-outlet management (create outlets, assign staff, inherit/override menu).
- Data backup & export (GDPR-style customer data export on request).

8. Non-Functional Requirements

Category	Requirement
Performance	Customer menu loads in $\leq 2s$ on 3G; order placement ACK in $\leq 500ms$ on good network.
Scalability	Architected to support 1 outlet $\rightarrow$ 500 outlets without re-engineering.
Availability	99.9% uptime; graceful degradation when components fail.
Offline resilience	KDS, Waiter, and Reception apps must queue actions offline and sync on reconnect — critical during internet outages common in the region.
Real-time sync	Order status changes propagate to all relevant surfaces within 1 second (WebSockets / push).
Security	HTTPS everywhere, encrypted at rest, role-based access, PIN/2FA for admin, rate limiting, OWASP Top 10 hardening.
Compliance	PCI-DSS SAQ-A via hosted payment iframes (no card data touches our servers). GDPR-style consent and data-deletion support. Local tax compliance (FBR for Pakistan: sales tax, POS integration if required).
Auditability	Full immutable audit log for all financial and permission-changing actions.
Accessibility	WCAG 2.1 AA on customer-facing surfaces.
Localization	i18n framework, RTL-ready (for Urdu/Arabic), multi-currency, per-outlet tax rules.
Maintainability	Modular monolith or service-oriented; feature flags for phased rollout.
Observability	Centralized logging, metrics, alerting, error tracking (Sentry-style).

9. Integrations

- **Payment gateways** — Coming soon
- **SMS** — Coming soon
- **WhatsApp Business API** — order confirmations, delivery tracking.
- **Email** — Nodemailer
- **Google Maps** — review prompt
- **Accounting** — custom CSV for local accountants.
- **Printers** — Local Printer
- **Hardware** — Coming Soon
- **Review platforms** — Google My Business API for review insights.

10. High-Level Data Model (Entities)

Outlet, User, Role, Permission, Table, Zone, Reservation, Customer, LoyaltyAccount, MenuCategory, MenuItem, Variant, Modifier, Combo, Ingredient, Recipe, RecipeItem, Supplier, PurchaseOrder, GoodsReceived, StockBatch, StockMovement, WastageLog, Order, OrderItem, OrderModifier, OrderStatusEvent, Payment, Refund, Bill, KitchenStation, Shift, Attendance, LeaveRequest, Promotion, Coupon, Campaign, Feedback, Review, DeliveryRider, DeliveryAssignment, AuditLog, NotificationTemplate.

(Full ERD to be produced in the technical design phase.)

11. Key User Flows (Summary)

Flow A — Dine-in Digital Order Scan QR → Browse menu → Add items & customize → Place order → Pay (or pay-at-counter) → Live tracking → Food served → Bill on phone → Feedback → Google review link.

Flow B — Kitchen Order Lifecycle Order received → Kitchen accepts + sets ETA → Station-wise prep → Items marked ready → Waiter notified → Served → Closed.

Flow C — Inventory Deduction Order placed → Recipe BOM fetched → Stock reduced per ingredient → Low-stock alert fires if par breached → Manager sees on dashboard → Raise PO → Receive stock → Stock replenished.

Flow D — Wastage Logging Item spoiled → Staff opens Waste Log → Selects item, qty, reason → Manager approves (if above threshold) → Stock deducted → Report updated.

Flow E — Reservation Customer books online → Deposit charged → Table blocked → Reminder SMS → Arrival → Check-in at reception → Deposit adjusted against bill.

12. MVP Scope vs Future Phases

12.1 MVP (Phase 1 — 3 to 4 months)

- Customer QR ordering (dine-in only), order tracking, digital bill, review prompt.

- Table & QR management (no reservations yet).
- Reception terminal with KOT print, bill print, basic payments (cash + 1 digital wallet + card via terminal).
- Kitchen Display System with accept / ETA / ready.
- Waiter app — notifications + status updates.
- Menu management with variants, modifiers, availability.
- Recipe/BOM linked auto inventory deduction.
- Basic inventory (stock in/out, low-stock alerts).
- Wastage log.
- Core reports (sales, menu performance, inventory, wastage).
- Single outlet, English + Urdu.

12.2 Phase 2 (Months 5–7)

- Reservations & waitlist.
- Takeaway & delivery modules + in-house rider management.
- Customer CRM + loyalty program.
- Promo codes, combos, dynamic pricing.
- Staff scheduling, attendance, payroll export.
- Full procurement (PO/GRN).
- Multi-outlet support.
- Advanced analytics & anomaly alerts.

12.3 Phase 3 (Months 8–12)

- Third-party delivery aggregator integrations.
- WhatsApp Business ordering.
- Marketing campaigns (bulk SMS/email).
- Menu engineering matrix and A/B testing.
- Franchise / multi-tenant mode.
- Mobile app for owners.
- AI features: demand forecasting, auto-reorder suggestions, menu price optimization, review sentiment analysis.

13. Technical Stack (Recommended)

Layer	Recommendation
Customer PWA	Next.js (typescript) + Tailwind; installable, offline-aware.
Staff apps (KDS, Waiter, Reception)	PWA

Backend	Node.js (with TypeScript).
Real-time	WebSockets (Socket.io)
Database	MongoDB (primary), Redis (cache, session, queues).
Search	Meilisearch/Elastic later.
Queue	BullMQ / Celery for background jobs (notifications, reports).
File storage	Cloudinary
Hosting	Vercel + Render + CDN (Cloudflare).
CI/CD	GitHub Actions; staging + production environments.
Monitoring	Sentry, Grafana + Prometheus, Logflare.
Auth	JWT + refresh tokens

14. Hardware Requirements (per outlet)

- **Reception** — 1× touchscreen POS (15"+), thermal receipt printer, cash drawer, card reader.
 - **Kitchen** — 1–2× rugged 10–13" tablets (wall-mount), 1× kitchen thermal printer per station.
 - **Waiters** — personal Android phones (BYOD acceptable) or shared 6" devices.
 - **Network** — business-grade Wi-Fi with backup 4G router (critical for offline resilience).
 - **Router/firewall** — segmented guest Wi-Fi vs POS network.
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15. Security & Compliance

- All customer payment flows via PCI-compliant hosted fields; card data never touches our servers.
- Passwords hashed (Argon2/bcrypt), 2FA for admin/manager roles.
- Role- and outlet-scoped access — a manager of Outlet A cannot see Outlet B data.
- Rate limiting, CAPTCHA on public endpoints (review submission, contact forms).
- Regular dependency scans, penetration testing pre-launch and annually.
- Data retention policy and customer data export/deletion on request.
- Encrypted backups, off-site, with tested restore procedure.
- Audit log of all privileged actions.